Selective Parameter Freezing for Cross-Dataset Paddy Disease Classification

Wallace F L Chin
School of Computing and Informatics
Universiti Teknologi Brunei
Bandar Seri Begawan, Brunei Darussalam
b20200347@student.utb.edu.bn

Wida S Suhaili
School of Computing and Informatics
Universiti Teknologi Brunei
Bandar Seri Begawan, Brunei Darussalam
wida.suhaili@utb.edu.bn

Daphne T C Lai
School of Digital Science
Universiti Brunei Darussalam
Bandar Seri Begawan, Brunei Darussalam
daphne.lai@ubd.edu.bn

***Abstract*— Accurate and timely diagnosis of paddy diseases is essential for reducing yield losses and supporting sustainable crop production. Although transfer learning has become the standard approach for plant disease classification, most existing studies adopt conventional full fine-tuning of pretrained models with limited investigation into whether alternative fine-tuning strategies improve cross-dataset generalisation. This work benchmarks representative convolutional and transformer architectures under cross-dataset evaluation to identify suitable backbone models, then investigates whether selective parameter freezing can improve generalisation by preserving transferable pretrained representations while allowing task-specific adaptation where required. DeiT-Small and ConvNeXt-Base were selected for evaluation under multiple freezing strategies, using public datasets for training and an independent local dataset for testing. Results show that the effects of freezing are architecture and component dependent. The best-performing configuration, freezing the depthwise convolution layers of ConvNeXt-Base, increased test accuracy at the peak validation-accuracy checkpoint from 61.5% to 63.8% and the best test accuracy observed across epochs from 64.5% to 69.2%, while reducing the number of trainable parameters. Qualitative analysis using Grad-CAM++ further indicates that appropriate freezing strategies preserve coherent attention over disease-relevant regions. These findings indicate that selective parameter freezing can improve cross-dataset performance while reducing the extent of parameter updates under the evaluated conditions.**

Keywords—Paddy Disease Classification, Rice Disease, Cross-Dataset Generalisation, Transfer Learning, Parameter Freezing

I. Introduction

Rice is a primary food source for nearly half of the global population, making reliable paddy disease diagnosis essential for reducing crop losses and supporting global food security [1], [2]. Recent advances in deep learning have enabled automated disease classification from images, with convolutional neural networks (CNNs) and Vision Transformers (ViTs) achieving high accuracy across a wide range of plant disease benchmarks [3]. These models are commonly developed through transfer learning, where networks pretrained on large-scale image datasets such as ImageNet are adapted to downstream agricultural tasks.

Despite their success, most existing studies evaluate transfer learning using randomly partitioned versions of the same dataset, where training and testing samples share similar imaging conditions. In practice, however, agricultural deployment often requires models to generalise across differences in background, illumination, viewpoint, acquisition devices, and disease presentation. Such distribution shifts frequently cause models that achieve excellent validation accuracy on curated benchmark datasets to exhibit significant performance degradation when evaluated on real-world field images [4], [5]. Improving cross-dataset generalisation therefore remains a critical challenge in practical plant disease classification.

Most existing studies employ full fine-tuning, allowing all pretrained parameters to adapt to the target dataset. While this maximises model flexibility, it may also overwrite generic visual representations learned during pretraining, particularly when the target dataset is relatively small or differs substantially from the deployment environment. Transfer learning literature suggests that lower layers of deep neural networks primarily encode generic visual primitives such as edges, textures, and simple shapes, whereas deeper layers capture increasingly task-specific semantic representations [6], [7]. This suggests that preserving transferable low-level representations while adapting higher-level features may provide a useful balance between knowledge retention and task adaptation. Although selective parameter freezing has been widely adopted as a practical transfer learning strategy, its effect on cross-dataset paddy disease classification, particularly across fundamentally different backbone architectures, remains insufficiently understood.

This work presents a systematic empirical study of selective parameter freezing for cross-dataset paddy disease classification. Representative CNN and ViT architectures are evaluated using depth- and component-based freezing strategies, with publicly available datasets used for training and an independent local field dataset for testing. This study compares freezing strategies across the two architectural families to investigate how parameter adaptation affects generalisation under distribution shift.

II. Related Work

A. Paddy Disease Classification

Early studies on paddy disease classification primarily employed transfer learning using convolutional neural networks (CNNs) such as ResNet and VGG, achieving high classification accuracy under controlled conditions [1]-[3]. More recently, attention mechanisms and ViTs have been increasingly adopted due to their ability to model long-range dependencies and capture fine-grained disease characteristics [3]. Subsequently, hybrid CNN-transformer architectures and attention-guided networks have further improved discrimination between visually similar disease classes [8]-[10].

While numerous studies report strong performance on benchmark paddy disease datasets, relatively few explicitly investigate cross-dataset generalisation. Notably, Yang et al. [9] reported a substantial performance drop when models trained on laboratory images were evaluated on independently collected field data, highlighting the difficulty of transferring learned representations across different acquisition environments. Similarly, Ahmad et al. [4] reported that models achieving over 99% accuracy under conventional train-test splits fell below 50% when evaluated on separate field images. These findings highlight the gap between benchmark performance and real-world generalisation, which remains comparatively underexplored in paddy disease classification.

B. Transfer Learning and Selective Fine-Tuning

Transfer learning has become the standard paradigm for plant disease classification, enabling models pretrained on large-scale datasets such as ImageNet to be adapted to agricultural tasks with limited labelled data. Two principal strategies are commonly employed: feature extraction, where the pretrained backbone is frozen and only the classification head is trained, and fine-tuning, where part or all of the pretrained network is updated using the target dataset [6], [7]. Partial fine-tuning provides an intermediate approach that retains selected pretrained representations while allowing task-specific adaptation.

Several studies have demonstrated that transfer learning performance depends not only on the pretrained model but also on how pretrained knowledge is utilised. Comparative studies on weight initialization showed that different transfer sources affect convergence and downstream performance, highlighting the importance of leveraging pretrained representations effectively [11]. Likewise, progressive fine-tuning strategies have demonstrated improved classification performance by gradually adapting pretrained representations rather than updating the entire network simultaneously [12]. Extending this line of research, Dong et al. [13] showed that the choice of fine-tuning paradigm significantly influences out-of-distribution plant disease recognition, with visual prompt tuning outperforming full fine-tuning in few-shot settings. Collectively, these studies indicate that the adaptation strategy itself plays a critical role in determining transfer learning performance.

CNNs and ViTs exhibit different transfer learning behaviour due to their architectural characteristics. CNNs incorporate strong inductive biases through local convolutions, making them comparatively data-efficient [14], [15], whereas ViTs rely more heavily on large-scale pretraining to learn spatial relationships through self-attention [15], [16]. Consequently, the effectiveness of parameter adaptation may differ between architectures. However, selective parameter freezing remains insufficiently studied under cross-dataset evaluation, where retaining transferable representations may be particularly important. This study addresses this gap by evaluating depth- and component-based freezing strategies across representative CNN and ViT backbones.

III. Methodology

This study follows a controlled comparative design in which all models share identical datasets, preprocessing pipeline, optimisation settings, and evaluation protocol. Selective parameter freezing strategy is the only experimental variable, allowing its effect on cross-dataset generalisation to be isolated. The experimental workflow is illustrated in Fig. 1. Images from publicly available datasets were first consolidated and manually preprocessed through the removal of edge cases and spatial refinement via cropping. ImageNet-pretrained backbone networks were then fine-tuned and benchmarked to identify representative CNN and ViT architectures for the subsequent freezing experiments. The resulting models were evaluated on an independently collected dataset to measure generalisation under distribution shift. Thus, all reported results reflect performance on an unseen data distribution rather than a conventional within-dataset split.

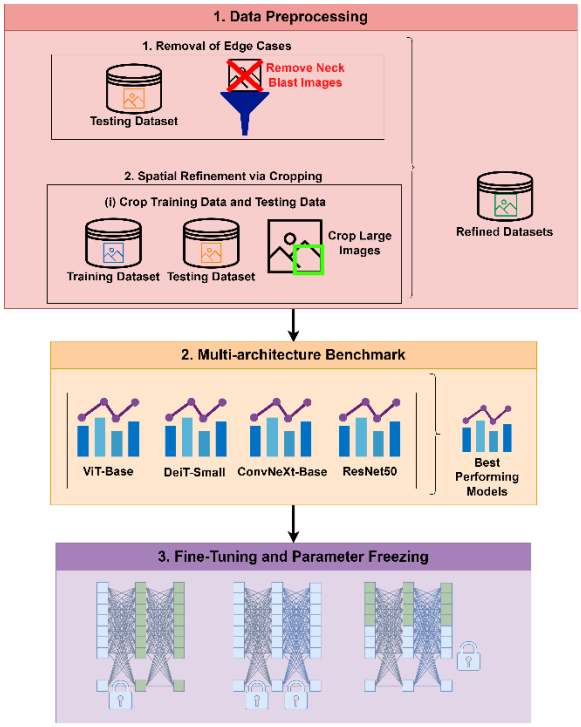


Fig. 1. Overall Methodology Overview

A. Datasets

The training dataset was constructed by consolidating seven publicly available paddy disease datasets, providing greater diversity in imaging conditions and disease presentation. The resulting dataset contains 20,744 labelled RGB images spanning eight disease and one normal category. The dataset includes controlled and field images, as well as leaf-level and plant-level views, capturing variation in backgrounds, imaging conditions, and spatial context. Table I summarises the constituent datasets and Fig. 2 provides sample images. A stratified 10% validation split, preserving the overall class distribution, was used to monitor overfitting during training.

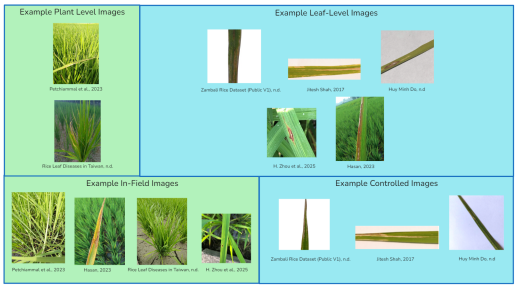


Fig. 2. Sample Images In The Training And Validation Datasets

TABLE I. SUMMARY OF DATASETS USED FOR TRAINING AND VALIDATION

Dataset	Bacterial Leaf Blight	Bacterial Leaf Streak	Bacterial Panicle Blight	Blast	Brown Spot	Dead Heart	Hispa	Normal	Sheath Blight	Notes
[17]	479	380	337	1738	965	1442	1594	1764	-	Labelled images only
[18]	-	-	-	779	523	-	565	1488	-	-
[19]	208	-	-	310	274*	-	225	187	288	*'Narrow Brown Spot' excluded
[20]	60	-	-	140*	95	-	-	309	-	*Mislabeled Brown Spot images in Blast folder excluded
[21]	40	-	-	-	40	-	-	-	-	-
[22]	-	-	-	519	92	-	-	316	-	-
[23]	-	-	-	1500*	1413	-	-	1500*	1174	*First 1500 images by filename selected
Total Images	**787**	**380**	**337**	**4986**	**3402**	**1442**	**2384**	**5564**	**1462**	**20744**

Evaluation was performed using an independent testing dataset comprising 758 images, including 619 images collected locally in Brunei and 139 supplementary images obtained from established agricultural repositories, specifically the International Rice Research Institute (IRRI) and the Center for Invasive Species and Ecosystem Health (invasive.org). The local images consisted primarily of field-collected samples, with additional controlled-environment images, while the supplementary repository images were included to improve representation of disease classes with limited local samples. To reduce the possibility of overlap with the training data, the complete testing dataset was screened for duplicate and near-duplicate images using the CleanVision near-duplicate detection library. As the testing dataset was collected independently of the training sources, it serves as an external evaluation set for assessing cross-dataset generalisation. Sample images from the testing dataset is shown in Fig. 3, while Fig. 4 depicts the class distribution of the testing dataset.

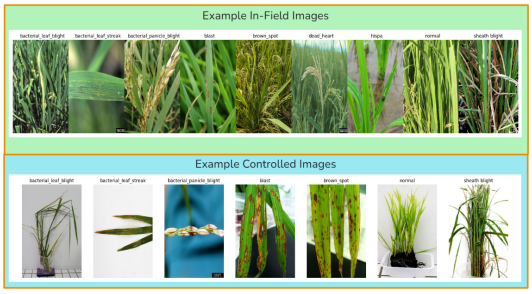


Fig. 3. Sample Images From Testing Dataset

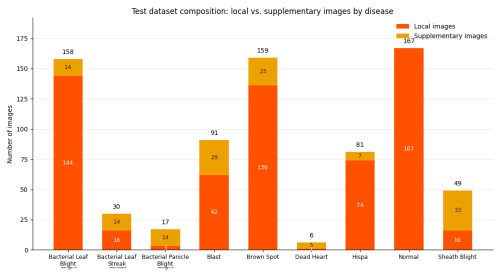


Fig. 4. Class Distribution Of Testing Dataset

B. *Data Preprocessing*

Before training, the consolidated dataset underwent limited manual preprocessing. Clearly mislabelled samples were removed, and high-resolution images with excessive background were manually cropped to better retain visible disease symptoms. Manual cropping was not applied to the independent testing images, and no disease-location information was used to guide this process. These preprocessing steps were performed once to produce the final dataset used throughout all experiments.

Training images were augmented using a pipeline consisting of random resized cropping (scale = 0.85–1.0, aspect ratio = 0.9–1.1), random horizontal flipping (0.5 probability), random vertical flipping (0.3 probability), and RandAugment (N = 2, M = 9). The RandAugment operation pool comprised autocontrast, histogram equalisation, brightness, contrast, saturation, and sharpness adjustment, together with translation and rotation transforms. Validation and test images were not augmented. All images were resized to 224 × 224 pixels and normalised using the ImageNet mean and standard deviation in accordance with the pretrained weights used.

C. *Training Configuration*

All models were initialised with ImageNet-pretrained weights and trained using the AdamW optimiser with a learning rate of 8×10^{-5}, weight decay of 5×10^{-4}, and a batch size of 32. Class imbalance was addressed using class-weighted cross-entropy loss, where weights were computed from the inverse class frequencies of the training set. A stratified 10% validation split was used for model selection, and training was terminated using early stopping with a patience of 12 epochs based on validation accuracy. All experiments were implemented in PyTorch using the timm model library and executed on an NVIDIA RTX 3060 (12 GB VRAM). Each experiment was conducted using three fixed random seeds (1337, 1338, and 1339) to account for stochastic variation, in which results are averaged.

D. *Pretrained Baseline Models*

Four ImageNet-pretrained architectures representing CNN and Transformer model families were initially evaluated using

identical training settings, as shown in Table II. Rather than performing an exhaustive architectural comparison, this stage was intended to identify representative CNN and ViT backbones for the subsequent experiments in Section III-E.

TABLE II. SUMMARY OF MODELS USED

Model Family	Architecture	Model Name	Pretrained Variant	No. of Parameters
Vision Transformer (**ViT**)	Vision Transformer	vit_base_patch16_224	vit_base_patch16_224.augreg2_in21k_ft_in1k	85.8M
Data-efficient Image Transformer (**DeiT**)	Vision Transformer	deit3_small_patch16_224	deit3_small_patch16_224.fb_in1k	21.7M
ConvNeXt	CNN	convnext_base	convnext_base.fb_in22k_ft_in1k	87.6M
ResNet	CNN	resnet50	resnet50.a1_in1k	23.5M

E. *Selective Parameter Freezing*

Transfer learning used ImageNet-pretrained weights, with a fully fine-tuned model established as the baseline for each backbone. Freezing experiments progressively restricted parameter updates while keeping all other training settings unchanged, isolating the effect of the freezing strategy.

Selective parameter freezing was investigated by restricting gradient updates to specific network portions while leaving the remainder trainable. Strategies were grouped into two categories: (1) Depth-based freezing, which freezes earlier network stages or transformer blocks to preserve generic representations while allowing deeper layers to adapt; and (2) Component-based freezing, which freezes specific architectural modules regardless of network depth to examine their contribution to transfer learning.

Because CNNs and Vision Transformers organise their representations differently, freezing strategies were defined according to the native components of each architecture rather than enforcing equivalent layer counts. This allows comparable adaptation strategies to be evaluated while respecting the structural differences between the two model families. The layer/component definitions for each freezing strategy are detailed in Tables IV and V.

IV. RESULTS AND ANALYSIS

A. *Experimental Evaluation*

Model performance was evaluated using validation and testing accuracy. Validation accuracy was used to monitor optimisation on the training domain and to select the checkpoint under the early-stopping procedure; accordingly, the testing accuracy at the validation-selected checkpoint was used as the primary measure of cross-dataset generalisation. For additional transparency, the maximum testing accuracy observed across epochs is also reported as a separate descriptive measure, but was not used for model, backbone, or freezing-strategy selection. This distinction is particularly relevant because the evaluated configurations differ substantially in parameter count and were trained using a shared learning rate and early-stopping patience. GradCAM++ visualisations at the earliest seed were used to qualitatively assess how selective parameter freezing influences the regions contributing to model predictions.

B. *Baseline Model Comparison*

TABLE III. RESULTS OF PRETRAINED BACKBONE ARCHITECTURE BENCHMARK

Model	Parameters	Val Accuracy	Test Accuracy	Test Accuracy (Best Epoch)
ViT-Base	85.8M	0.9245	0.4126	0.4597
DeiT-Small	21.7M	0.9362	0.4464	0.4937
ConvNeXt-Base	**87.6M**	**0.9577**	**0.6147**	**0.6447**
ResNet-50	23.5M	0.9398	0.5193	0.5312

Table III presents the baseline performance of four ImageNet-pretrained architectures under the proposed cross-dataset evaluation protocol. Although all models achieved high validation accuracy (92.5–95.8%), test accuracy was substantially lower at the peak validation-accuracy checkpoint, ranging from 41.3% to 61.5%, demonstrating the challenge of cross-dataset generalisation. ConvNeXt-Base achieved the highest test accuracy (61.5%), followed by ResNet-50 (51.9%), while the ViT-based models showed greater performance degradation despite competitive validation accuracy. The peak test accuracy observed across epochs was higher for all models, ranging from 46.0% to 64.5%, suggesting that the epoch with the highest validation accuracy did not necessarily correspond to the epoch with the highest external-test accuracy. This divergence indicates that validation performance may not fully reflect cross-dataset generalisation, particularly under the observed distribution shift. However, these values are reported separately as descriptive results and were not used for model selection. These results suggest that the CNN backbones generalised more effectively under the evaluated distribution shift. Consequently, ConvNeXt-Base and DeiT-Small were selected as representative CNN and ViT backbones for the subsequent freezing experiments.

The quantitative results were further supported by GradCAM++ visualisations in Fig. 5. CNN-based architectures generally concentrated their activations on symptomatic regions, whereas transformer-based models frequently attended to surrounding background structures.

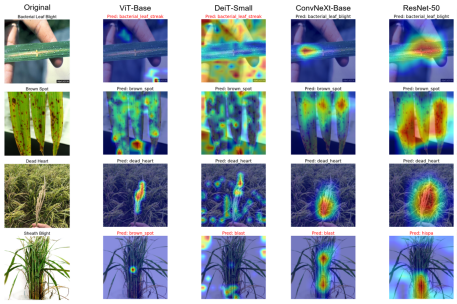


Fig. 5. Gradcam++ Activations For Benchmark Architectures

C. *Selective Parameter Freezing Results*

Tables IV and V summarise the effect of selective parameter freezing on DeiT-Small and ConvNeXt-Base, respectively. Across both architectures, the results show that the proportion of frozen parameters alone does not consistently determine performance; the location and type of the frozen components also influence the outcome. Differences between test accuracy at the peak validation-accuracy checkpoint and the best test accuracy across epochs

further indicate some divergence between validation and external-test behaviour. Across both architectures, freezing strategies that preserved earlier pretrained representations generally produced stronger cross-dataset performance than full fine-tuning or freezing deeper network components.

TABLE IV. RESULTS OF VARIOUS FREEZING STRATEGIES ON DEIT-SMALL

Model	**DeiT-Small** (12 transformer encoder blocks total)				
Freezing Strategy	**Strategy Type**	**Trainable/ Total Parameters (% frozen)**	**Val Accuracy**	**Test Accuracy**	**Test Accuracy (Best Epoch)**
Baseline	Full fine-tuning	21.7M/21.7M (0.0%)	0.9362	0.4464	0.4937
Freeze First ½ Blocks	Depth-Based (Blocks)	11.0M/21.7M (49.3%)	0.9408	0.4315	0.5608
Freeze First ¼ Blocks	Depth-Based (Blocks)	16.4M/21.7M (24.4%)	**0.9429**	**0.4612**	**0.5710**
Freeze First ¾ Blocks	Depth-Based (Blocks)	5.7M /21.7M (32.7%)	0.9362	0.4451	0.5049
Freeze Last ½ Blocks	Depth-Based (Blocks)	11.0M/21.7M (49.3%)	0.9406	0.4118	0.4512
Freeze All Attention	Parts-Based	14.6M/21.7M (32.7%)	0.9325	0.4272	0.5059

TABLE V. RESULTS OF VARIOUS FREEZING STRATEGIES ON CONVNEXT-BASE

Model	**ConvNeXt-Base** (4 Stages with 36 ConvNeXt Blocks total)				
Freezing Strategy	**Strategy Type**	**Trainable/Total Parameters (% frozen)**	**Val Accuracy**	**Test Accuracy**	**Test Accuracy (Best Epoch)**
Baseline	Full fine-tuning	87.6M/87.6M (0.0%)	0.9577	0.6147	0.6447
Freeze Stage 0-1	Depth-Based (Stages)	85.4M/87.6M (2.5%)	0.9581	0.5869	0.6386
Freeze Stage 0-2	Depth-Based (Stages)	27.5M/87.6M (68.6%)	0.9462	0.4820	0.5765
Freeze Stage 2-3	Depth-Based (Stages)	2.2M / 87.6M (97.5%)	0.9365	0.4659	0.5396
Freeze All Depthwise Conv2d	Parts-Based	86.7M/87.6M (1.0%)	**0.9609**	**0.6483**	**0.6915**
Freeze All Conv2d	Parts-Based	81.9M/87.6M (6.5%)	0.9571	0.5799	0.6491
Freeze First ⅔ Blocks	Depth-Based (Blocks)	47.1M / 87.6M (46.2%)	0.9419	0.5730	0.6515

For DeiT-Small, freezing the first quarter of transformer blocks produced the best overall performance, improving testing accuracy from 0.4464 to 0.4612 and peak test accuracy from 0.4937 to 0.5710. Freezing the first half of the network also improved generalisation, whereas freezing later transformer blocks consistently reduced performance.

A similar trend was observed for ConvNeXt-Base. The best performance was achieved by freezing all depthwise convolution layers, increasing testing accuracy from 0.6147 to 0.6483 and peak test accuracy from 0.6447 to 0.6915. Moderate freezing of the early network stages also improved performance relative to full fine-tuning, whereas freezing later stages or excessive portions of the network consistently reduced testing accuracy.

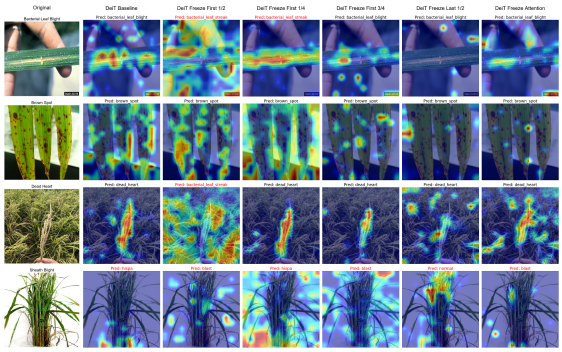


Fig. 6. Gradcam++ Activations Following Freezing Strategies On Deit-Small

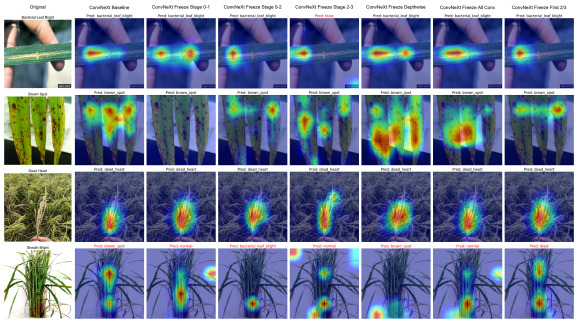


Fig. 7. Gradcam++ Activations Following Freezing Strategies On ConvNeXt-Base

GradCAM++ visualisations in Figs. 6 and 7 provides qualitative insight into how selective parameter freezing influenced feature localisation. For DeiT-Small, activation patterns varied substantially across different freezing strategies, indicating that feature localisation was highly sensitive to the choice of frozen transformer blocks. In contrast, ConvNeXt-Base produced comparatively consistent activation patterns across freezing configurations, suggesting that its learned representations were more stable despite changes to parameter adaptation.

Overall, the experimental results demonstrate that selective parameter freezing can improve cross-dataset generalisation when applied to transferable pretrained representations while preserving the adaptability of higher-level task-specific features. Although the optimal freezing strategy differs between CNN and Vision Transformer architectures, both exhibit the same underlying trend: preserving early representations is generally more beneficial than freezing later semantic layers. These findings suggest that selective parameter freezing provides a relatively simple transfer-learning strategy that reduces the number of parameters requiring optimisation.

V. CONCLUSION AND FUTURE WORK

This paper investigated selective parameter freezing as a parameter-efficient transfer learning strategy for cross-dataset paddy disease classification. The results indicate that selected

freezing configurations can improve cross-dataset performance relative to full fine-tuning, with effects that vary by architecture and freezing location. The best-performing configuration, achieved by freezing the depthwise convolution layers of ConvNeXt-Base, improved testing accuracy from 61.5% to 64.8% and peak testing accuracy from 64.5% to 69.2% while reducing the number of trainable parameters. These findings suggest that selective freezing can reduce optimisation requirements while improving performance under the evaluated cross-dataset conditions. The substantial gap between validation accuracy on the pooled public datasets and independent test performance indicates a pronounced generalisation gap, which may reflect both overfitting and distributional differences that were not fully disentangled in this study. Some Grad-CAM++ visualisations also showed attention to background vegetation and field clutter rather than lesion sites, suggesting possible reliance on non-disease-specific visual cues.

Several limitations should be considered. First, the study focuses on paddy disease classification and does not establish whether the findings generalise to other agricultural or visual classification tasks. Second, all backbones use ImageNet-pretrained weights, so the observed effects may depend partly on the pretraining source. Third, cross-dataset generalisation is evaluated primarily against an independently collected target domain from Brunei. Although supplementary repository images improved representation of underrepresented classes, they do not provide equivalent coverage across multiple geographic target domains. The findings should therefore be interpreted within the evaluated conditions.

Future work should examine selective freezing across more architectures, datasets, and agricultural classification tasks. Analysing representation stability across domains could clarify why particular layers benefit from being preserved during adaptation. Given the observed background activation, advanced regularization, foreground masking, disease-region cropping, or background manipulation could be evaluated to reduce reliance on background cues. More adaptive parameter-efficient approaches, including learnable or data-dependent freezing, could also be investigated under distribution shift. Finally, combining selective freezing with parameter-efficient adaptation modules or domain adaptation may determine whether complementary adaptation strategies provide further benefits on independently collected target-domain data.

ACKNOWLEDGMENT

The authors express their appreciation to the Brunei Department of Agriculture and Agrifood and Plant Pathology Unit for their assistance and contribution to this work.

REFERENCES

[1] R. B. R. Firdaus, M. Leong Tan, S. R. Rahmat, and M. Senevi Gunaratne, ‘Paddy, rice and food security in Malaysia: A review of climate change impacts’, Cogent Soc. Sci., vol. 6, no. 1, p. 1818373, Jan. 2020, doi: 10.1080/23311886.2020.1818373.

[2] A. Islam, R. Islam, S. Haque, S. Islam, and M. A. I. Khan, ‘Rice leaf disease recognition using local threshold based segmentation and deep CNN’, Int J Intell Syst Appl, vol. 13, no. 5, pp. 35–45, 2021.

[3] W. F. L. Chin and W. S. Suhaili, ‘Paddy Disease Detection through Vision Transformers’, IOP Conf. Ser. Earth Environ. Sci., vol. 1426, no. 1, p. 012003, Dec. 2024, doi: 10.1088/1755-1315/1426/1/012003.

[4] A. Ahmad, A. E. Gamal, and D. Saraswat, ‘Toward Generalization of Deep Learning-Based Plant Disease Identification Under Controlled and Field Conditions’, IEEE Access, vol. 11, pp. 9042–9057, 2023, doi: 10.1109/ACCESS.2023.3240100.

[5] Y. Lebrini and A. Ayerdi Gotor, ‘Crops Disease Detection, from Leaves to Field: What We Can Expect from Artificial Intelligence’, Agronomy, vol. 14, no. 11, Art. no. 11, Nov. 2024, doi: 10.3390/agronomy14112719.

[6] M. A. Morid, A. Borjali, and G. Del Fiol, ‘A scoping review of transfer learning research on medical image analysis using ImageNet’, Comput. Biol. Med., vol. 128, p. 104115, Jan. 2021, doi: 10.1016/j.compbiomed.2020.104115.

[7] C. Tan, F. Sun, T. Kong, W. Zhang, C. Yang, and C. Liu, ‘A Survey on Deep Transfer Learning’, Aug. 06, 2018, arXiv: arXiv:1808.01974. Accessed: Jan. 13, 2024. [Online]. Available: http://arxiv.org/abs/1808.01974

[8] P. S. Thakur, P. Khanna, T. Sheorey, and A. Ojha, ‘Explainable vision transformer enabled convolutional neural network for plant disease identification: PlantXViT’, Jul. 16, 2022, arXiv: arXiv:2207.07919. doi: 10.48550/arXiv.2207.07919.

[9] Y. Yang, G. Jiao, J. Liu, W. Zhao, and J. Zheng, ‘A lightweight rice disease identification network based on attention mechanism and dynamic convolution’, Ecol. Inform., vol. 78, p. 102320, Dec. 2023, doi: 10.1016/j.ecoinf.2023.102320.

[10] A. A. Salamai, N. Ajabnoor, W. E. Khalid, M. M. Ali, and A. A. Murayr, ‘Lesion-aware visual transformer network for Paddy diseases detection in precision agriculture’, Eur. J. Agron., vol. 148, p. 126884, Aug. 2023, doi: 10.1016/j.eja.2023.126884.

[11] D. Sinanc Terzi, ‘Effect of different weight initialization strategies on transfer learning for plant disease detection’, Plant Pathol., vol. 73, no. 9, pp. 2325–2343, 2024, doi: 10.1111/ppa.13997.

[12] S. Acharya, T. Kar, S. Pal, P. K. Patra, and U. C. Samal, ‘Plant disease identification through Explainable AI: A deep learning architecture using fine-tuned EfficientNet for sustainable agriculture’, Frankl. Open, vol. 16, p. 100707, Sep. 2026, doi: 10.1016/j.fraope.2026.100707.

[13] J. Dong, A. Fuentes, H. Zhou, Y. Jeong, S. Yoon, and D. S. Park, ‘The impact of fine-tuning paradigms on unknown plant diseases recognition’, Sci. Rep., vol. 14, no. 1, p. 17900, Aug. 2024, doi: 10.1038/s41598-024-66958-2.

[14] J. Wang, Q. Xu, B. Jiang, B. Luo, and J. Tang, ‘Multi-Granularity Part Sampling Attention for Fine-Grained Visual Classification’, IEEE Trans. Image Process., vol. 33, pp. 4529–4542, 2024, doi: 10.1109/TIP.2024.3441813.

[15] M. Goldblum et al., ‘Battle of the Backbones: A Large-Scale Comparison of Pretrained Models across Computer Vision Tasks’, Nov. 20, 2023, arXiv: arXiv:2310.19909. doi: 10.48550/arXiv.2310.19909.

[16] R. Courant, M. Edberg, N. Dufour, and V. Kalogeiton, ‘Transformers and Visual Transformers’, in Machine Learning for Brain Disorders, O. Colliot, Ed., New York, NY: Springer US, 2023, pp. 193–229. doi: 10.1007/978-1-0716-3195-9_6.

[17] Petchiammal, B. Kiruba, Murugan, and P. Arjunan, ‘Paddy doctor: A visual image dataset for automated paddy disease classification and benchmarking’, presented at the Proceedings of the 6th Joint International Conference on Data Science & Management of Data (10th ACM IKDD CODS and 28th COMAD), 2023, pp. 203–207.

[18] Huy Minh Do, ‘Rice Diseases’. [Online]. Available: https://www.kaggle.com/datasets/minhhuy2810/rice-diseases-image-dataset

[19] M. Hasan, ‘Rice Leaf Bacterial and Fungal Disease Dataset’, vol. 1, Nov. 2023, doi: 10.17632/hx6f852hw4.1.

[20] ‘Zambali Rice Dataset (Public V1)’. Accessed: Aug. 10, 2025. [Online]. Available: https://www.kaggle.com/datasets/gettingintoml/zambali-rice-dataset-v3-1

[21] H. P. Jitesh Shah, ‘Rice Leaf Diseases’. UCI Machine Learning Repository, 2017. doi: 10.24432/C5R013.

[22] ‘Rice leaf diseases in Taiwan’. Accessed: Feb. 12, 2026. [Online]. Available: https://www.kaggle.com/datasets/raingel/rice-leaf-disease

[23] H. Zhou, D. Cai, L. Lin, D. Huang, and B.-M. Wu, ‘Recognition of multi-symptomatic rice leaf blast in dual scenarios by using convolutional neural networks’, Smart Agric. Technol., vol. 11, p. 100867, Aug. 2025, doi: 10.1016/j.atech.2025.100867.